

Dams mark drought vulnerability rather than
buffer it, and storage cannot offset a falling water
supply in Chile

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Abstract

Expanding reservoir storage is the reflexive policy response to intensifying drought, yet whether dams build resilience or merely mark where vulnerability already concentrates remains untested under sustained drying. We treat reservoir presence as a basin-level intervention across central Chile — 26 monitored reservoirs spanning the 2010–present megadrought — and estimate its effect on drought propagation and water-dependent land use using a national matched design (entropy balancing within Köppen × aridity strata; 21 of 24 treated basins retained, effective control sample 91), doubly-robust matched ATT, forcing-conditioned difference-in-differences, and randomization inference. Across every operational pathway — meteorological-to-hydrological drought transmission, evapotranspiration buffering, and irrigated-area expansion — the treatment effect brackets zero once siting is matched out. A within-basin placebo comparing regulated downstream reaches against physically unregulated upstream reaches of the same dammed basins (15 of 21 basins) is the cleanest within-design falsification: the apparent national buffering signal (slope gap -0.18) is as strong upstream as downstream and collapses to -0.03 within climate region ($p_{\text{perm}} = 0.39$) — the signature of an aridity-gradient confound, not regulation. Refill

efficiency is undiminished while the whole storage band drifts downward, a pattern consistent with declining water supply rather than a loss of storage or buffering capacity. Reservoirs in Chile thus appear to mark pre-existing, siting-determined vulnerability rather than make resilience; once siting is accounted for, the operational case for storage expansion as drought protection is not supported by these data.

Keywords: reservoir effect, drought propagation, socio-hydrology, causal inference, megadrought, Chile

1 Introduction

Reservoirs are the dominant structural response to drought: when water shortages inflict economic damage, public pressure tends to drive the expansion of storage to raise water availability, which in the short term lowers the frequency, severity and duration of shortage¹. Yet a socio-hydrological perspective warns that the same infrastructure can be self-defeating over the planning horizon of a dam (20–30 years) through two coupled feedbacks. In the *supply–demand cycle*, greater supply enables agricultural, urban or industrial expansion, so demand grows faster than projected from socio-economic trends alone and quickly offsets the initial benefit — a rebound, or Jevons, effect¹. In the *reservoir effect* — the drought analogue of the levee effect — prolonged reliance on abundant stored water erodes the incentive to adapt, deepening dependence on infrastructure and amplifying the damage when a drought finally exceeds storage capacity¹. Global accounting is consistent with this view: water demand has outpaced storage capacity in recent decades, and local water histories from Las Vegas, Melbourne and Athens show consumption rising to meet, and then exceed, each new increment of supply¹, and the storage returns of new dam construction are themselves diminishing worldwide². Critically, the rebound means the enhanced damage from over-reliance need not be offset by a lower frequency of shortage, making the safe-development paradox potentially *more* dangerous for drought than for floods¹. Whether these

feedbacks dominate in practice, or are outweighed by reservoirs' protective function, 093
 remains actively debated³. 094
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 The short-term, hydrological side of this ledger is well documented. As drought 097
 propagates from precipitation deficit through soil moisture to streamflow, modulated by 098
 catchment characteristics⁴⁻⁶, comparisons of regulated against naturalized streamflow 099
 show that reservoir operation reduces the duration and magnitude of short-term (1-3 100
 month) hydrological drought downstream and lengthens the lag between meteorological 101
 and hydrological drought^{7;8}, and that reservoir management can damp rapid drought-to- 102
 flood transitions, particularly in snowmelt-driven catchments⁹. But the same regulation 103
 can redistribute scarcity in space and time¹⁰: in the transboundary Tagus, operation 104
 of the Alcántara dam lengthened and intensified droughts *downstream* of the reservoir 105
 relative to the pre-dam period¹¹. The long-term, demand side is far less constrained 106
 empirically. Irrigation modernization that nominally "saves" water often expands 107
 irrigated area or shifts farmers toward higher-value, more water-intensive crops — the 108
 efficiency paradox — raising consumptive use and basin-scale vulnerability rather than 109
 lowering it^{12;13}. Whether reservoirs set the same trap remains largely untested. 110
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 Chile is an acute testbed. Since 2010 central Chile (30–38° S) has endured a *megadrought*: 112
 an uninterrupted run of dry years with mean rainfall deficits of 25–45 %, punctuated 113
 by "hyperdrought" winters in 2019 and 2021 with deficits exceeding 75 %^{14;15}. The 114
 event is unprecedented over the instrumental record and has no clear analogue in 115
 millennial tree-ring reconstructions; roughly one-quarter to one-third of its intensity 116
 is attributable to anthropogenic forcing — in line with warming-driven increases in 117
 drought risk documented globally^{16;17} — the remainder to Pacific decadal variability 118
 and long-term regional drying^{14;18-20}. The signal amplifies as it propagates: a ~30 % 119
 reduction in Andean snow-water equivalent²¹ — part of intensifying snow drought 120
 worldwide^{22;23} — translates into streamflow declines of 20–70 % (locally up to 90 %), 121
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139 with the streamflow deficit in arid sectors often twice the rainfall deficit, and reservoirs
 140 and groundwater drawn to historical lows^{14;24} — part of a global pattern in which
 141 most reservoirs now fail to refill under intensifying drought²⁵. Demand, however, has
 142 scarcely relented — infrastructure and intensive groundwater pumping have held human
 143 withdrawals near pre-drought levels, masking a “Day Zero” that surface water alone
 144 could not avert^{15;24} — at the cost of aquifer overexploitation now well documented in
 145 central Chile^{26–28}. Mounting evidence suggests the region is crossing from episodic
 146 drought into structural aridification, with long-term declines in water supply — and
 147 rising atmospheric evaporative demand — driving land-cover change^{29–32}.
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 149 These dynamics are inseparable from Chile’s water governance. The 1981 Water Code,
 150 the paradigm case of market-based water management, made water-use rights tradable,
 151 perpetual private property separable from land^{33;34}. Rights were granted on the
 152 stationarity assumption that historical flows would persist, producing over-allocated
 153 “paper rights” in many basins, while management was delegated largely to private
 154 water-user organizations^{35–38}. Because the state cannot readily curtail existing private
 155 rights during scarcity, the policy reflex has been supply-side: a national programme
 156 of new dams and inter-basin transfers^{15;35}. The same institutional setting rewarded
 157 conversion to high-value, water-demanding export crops — orchards and vineyards —
 158 concentrating water in agro-export firms and locking in consumptive demand even as
 159 supply contracted^{34;39}. Yet most farmers still meet scarcity with short-term, reactive
 160 measures rather than durable adaptation, leaving long-run vulnerability largely intact¹³.
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 162 Chile thus exhibits, in compressed form, every ingredient of the reservoir effect: an
 163 extreme and persistent forcing, a supply-side infrastructure response, market incentives
 164 for demand expansion, and weak adaptive feedback.
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 166 Despite its prominence, the reservoir effect for drought has been described far more
 167 than it has been *measured*. The framework’s authors note that we lack the datasets and
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analytical tools to quantify these social–technical–hydrological feedbacks, and — above
all — defensible counterfactuals: what a drought’s impact *would have been* in a basin
had the reservoir not been built¹. Existing work compounds the difficulty. Drought-
propagation studies remain mostly temporal, leaving the spatial footprint of upstream
regulation on downstream drought poorly resolved⁴⁰, and quantitative comparisons
of drought resistance across dams are scarce⁴¹. The core obstacle is identification:
reservoirs are not sited at random but built where irrigation demand and aridity already
concentrate, so dammed basins differ systematically from undammed ones before any
drought strikes. Basin-mean greenness, raw before–after contrasts, or calendar-time
comparisons cannot separate a genuine reservoir effect from this siting confound.

Here we ask whether reservoirs build resilience or merely postpone and concentrate
vulnerability during prolonged drought. We treat the presence of a major reservoir as a
basin-level intervention and construct the missing counterfactual by pairing a matched
dammed-versus-undammed design with a forcing-conditioned estimand — comparing
how dammed and hydroclimatically comparable undammed basins respond to the *same*
drought forcing. This design targets the long-term demand-side feedback that prior,
hydrology-only analyses leave open, and turns the socio-hydrological reservoir-effect
hypothesis into a falsifiable, causally framed test in the setting where it matters most.

2 Methods

2.1 Study area and design

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning
hyper-arid, Mediterranean, and humid-temperate climates, over 2005–2024 — the
window in which reservoir storage, meteorological forcing, and satellite outcomes jointly
exist (`config/study_period.yml`). The design is a matched comparison of **dammed**
versus undammed sub-watersheds, with reservoir presence treated as a basin-level

intervention. Because 18 of 24 monitored reservoirs were commissioned before the storage record begins, treatment is effectively time-invariant over the window and a staggered-adoption (commissioning) event study is infeasible; identification therefore rests on a matched cross-section combined with conditioning on meteorological forcing rather than on calendar time.

The central estimand is the operational effect of a reservoir on drought outcomes *conditional on the covariates that govern where reservoirs are sited* — that is, whether dammed basins respond to the same drought forcing differently from hydroclimatically comparable undammed basins. This isolates reservoir *operation* from reservoir *siting*¹.

2.2 Spatial units and treatment assignment

The analysis grain is the DGA BNA **sub-watershed (subcuenca)** (467 units), chosen over the coarser cuenca grain (101 units) for power and common support: reservoirs sit in Chile’s largest main-stem basins, and the finer grain supplies far more controls inside the treated size range (`docs/design/grain-selection.md`). A subcuenca is **treated** if it contains 1 of the 26 monitored reservoirs by point-in-polygon assignment, giving **24 treated units**. An untreated subcuenca that lies within a cuenca containing a reservoir is flagged as **contaminated** — it shares the regulated main stem and is not a clean counterfactual — and removed (45 units), leaving **398 clean controls**. This contamination rule defuses the most direct downstream-spillover (SUTVA) violation; residual adjacency and shared-aquifer spillovers between neighbouring subcuencas are carried as an identification caveat and checked at cuenca grain.

2.3 Data

Reservoir storage and treatment. Monthly storage for 26 monitored reservoirs (2005-01 to 2026-04) with a per-reservoir capacity column (`max_level_hm3`); storage and capacity share units (hm^3), so percent-of-capacity ($100 \times \text{level} /$

`max_level_hm3`) is the cross-reservoir-comparable state variable. Static attributes
(commissioning year, use, size class) come from the national dam registry.

Meteorological forcing. Standardized drought indices — SPI, SPEI, and EDDI — derived from CHIRPS precipitation and CHIRTS temperature at timescales of 1/3/6/12/24/36 months, standardized on the 1991–2026 baseline (`config/variables.yml`). SPEI⁴² at 12 months is the primary exogenous dose; the suitability of CHIRPS-derived standardized indices for Chilean drought monitoring is established, including ENSO-modulated drought variability across the country^{43–45}. We use the 12-month accumulation because it matches the annual, snowmelt-fed hydrological cycle of central Chile — winter precipitation stored as snow and released through the growing season — and so captures the inter-seasonal carryover that irrigation reservoirs are built to bridge, at the scale at which multi-year vulnerability (rather than sub-seasonal operation) is expressed. By construction this does not resolve short-term (1–3 month) operational buffering, which reservoirs are documented to provide^{7;8} and which we flag as a boundary of our conclusions (Discussion). Conditioning on forcing — estimating effects on the deficit-to-impact *slope* rather than on levels — is how we break the collinearity between the 2010-onset megadrought and the treatment clock.

Hydrological drought. Daily streamflow from the CR2 network (2000–2020) is aggregated to the Standardized Streamflow Index at 12 months [SSI-12; (**author?**)⁴⁶; (**author?**)⁴⁷], the regulated-flow analogue of SPEI-12, used as the outcome in the within-basin placebo (below). Gauges are classified as upstream or downstream of each dam from an SRTM digital elevation model.

Ecological and land-use outcomes. (i) annual irrigated/cropland area from Map-Biomas Chile Collection 2 (30 m, 1999–2024); (ii) actual evapotranspiration from MODIS (8-day, ~500 m, 2001–2024), for whole basins and for an orchard stratum from

the CIREN/ODEPA cadastre; (iii) standardized vegetation and productivity anomalies — monthly z-score accumulated NDVI (zcNDVI-6), annual z-score NPP (zNPP), and the standardized actual-ET anomaly (SETI) — from MODIS over 2000–2024; the zcNDVI productivity proxy is validated for Chilean agricultural drought, and the ecological response of Chilean vegetation to the megadrought is well characterized^{48–50}.

Matching covariates. Baseline aridity (long-term mean annual P/PET over the 1991–2020 WMO normal), watershed area, mean elevation (SRTM), and Köppen–Geiger climate class, extracted per subcuenca by zonal statistics.

2.4 Matched control construction

We estimate the average treatment effect on the treated (ATT) using **entropy balancing** (`WeightIt, method = "ebal"`) exact-matched within Köppen main group, balancing **log(area)** and **mean elevation** (`docs/design/matched-controls.md`). Balancing on size and elevation rather than latitude is deliberate: latitude and the climate class fight each other across the Andes–Patagonia span, and substituting SRTM elevation raised the effective control sample size from 9 to 91 while balancing latitude for free.

The procedure retains **21 of 24 treated subcuencas** (three high-Andes units fall outside local elevation common support) and achieves essentially perfect balance — standardized mean differences for `log(area)` and `elevation` fall from 1.18 / 0.37 to 0.000, variance ratios 1 — with an **effective control sample size 91** out of 244 in-window controls. Baseline aridity is a genuine confound (dammed basins are markedly more arid: median P/PET 0.26 vs 0.80) but is *not* used as a balance target, because hard-balancing it collapses the ESS from 91 to 19; instead the residual aridity imbalance (SMD 0.17, above the 0.1 target) is absorbed by doubly-robust regression adjustment. Because the matched set does not fully balance aridity, this adjustment is *load-bearing*.

rather than cosmetic, and the estimates can be sensitive to its functional form; we
therefore stress-test the aridity-adjusted estimates with a quadratic aridity term and
report the curvature-robust values where feasible (see Results). Robustness to the
matching method is assessed with coarsened exact matching (CEM) and 1:2 nearest-
neighbour (Mahalanobis) matching on the identical common-support sample; entropy
balancing and NN agree on all 21 treated, while CEM additionally prunes the 4 most
extreme large/high basins, so all treatment effects are reported with and without those
units.

2.5 Causal estimands and estimators

Doubly-robust matched ATT. Each outcome is estimated three ways on the
same sample — weighting-only (ebal-weighted difference in means), regression-only
(covariate-adjusted g-computation), and the headline **doubly-robust** estimator (ebal
weights *and* an outcome model in log(area), elevation, and log(aridity)) — so functional-
form sensitivity is visible. Continuous covariates are centred at the treated mean so the
treatment coefficient reads as the ATT; standard errors are HC3-robust (ebal weights
treated as fixed, the standard mildly anti-conservative simplification).

Forcing-conditioned transmission slope. To remove megadrought *exposure* (con-
centrated in exactly the arid basins where reservoirs sit), the outcome for the DR
machinery is the per-basin **transmission slope** — the regression of an annual out-
come (e.g. zNPP) on annual SPEI-12 over 2005–2024 — a standardized, cross-basin
drought-transmission elasticity. The vulnerability hypothesis predicts a *steeper* slope
(more impact per unit deficit) in dammed basins.

Forcing-interacted difference-in-differences. On the observed area and ET panels
we fit

$$y_{it} = \beta (\text{dammed}_i \times \text{SPEI}_{it}) + \gamma \text{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

with subcuena (α_i) and year (τ_t) fixed effects and entropy-balancing weights. Because treatment is time-invariant, the dose is meteorological SPEI rather than calendar year: the year fixed effects absorb the common megadrought shock, and β is the differential deficit-to-impact slope between dammed and matched-control basins — robust to the siting-in-arid-central-Chile confound, which biases levels far more than slopes. The vulnerability hypothesis predicts $\beta > 0$.

Equivalence testing and minimum detectable effect. Because the headline results are nulls, we move beyond failure-to-reject and bound the operational effect. For each null we report a two one-sided test (TOST): the effect is declared equivalent to zero if its 90% confidence interval lies within a negligible region of $\pm 25\%$ of the baseline (untreated) transmission slope. We are not aware of an established hydrological or operational standard defining a “negligible” reservoir effect on drought transmission, so we anchor the margin in the granularity of operational drought classification rather than choosing it arbitrarily. Standardized drought indices are managed in discrete severity categories each 0.5 standardized units wide^{42;51}, over the observed range of meteorological deficits a change of $< 25\%$ in the deficit-to-impact slope (0.13 SSI units on the 0.53 baseline) alters the propagated streamflow-drought index by less than a single severity category, i.e. it cannot move a basin across an operational drought-severity threshold and is therefore marginal for drought management. We adopt $\pm 25\%$ as this deliberately strict, pre-specified margin and, because the equivalence verdict is nonetheless sensitive to the bound, also report the threshold-free MDE so that readers may apply any margin they consider material. The qualitative conclusion (only large effects are excluded; no outcome is equivalent at $\pm 25\%$) does not depend on the exact margin. The minimum detectable effect (at 80% power, two-sided $\alpha = 0.05$) and the

equivalence interval are built from the **standard deviation of the permutation-null distribution**, not the analytic standard error, since cluster-robust SEs over-reject at ~21 clusters (`src/R/causal/equivalence.R`). To confirm the null is informative rather than underpowered, we additionally run a **positive control**: a known buffering slope is injected into the treated units and the full estimator and randomization inference are re-run, verifying that the design recovers an effect of policy-relevant size and correctly flags none when none is injected.

Siting-confound decomposition. To quantify how much of any apparent buffering is selection, we fit a ladder on the same panel: (1) a naive pooled slope gap (no weights, no fixed effects — what a conventional dammed-vs-undammed study reports); (2) unit + time fixed effects, unweighted; (3) the design estimator (ebal weights + fixed effects). The shrinkage from (1) to (3) is the siting confound; the residual at (3) is the regulation estimate.

Within-basin upstream/downstream placebo. The primary selection-versus-effect discriminator contrasts the SPEI-12→SSI-12 transmission slope at regulated **downstream** reaches against physically unregulated **upstream** reaches of the *same* dammed basins, which share climate, geology, and selection history. A genuine regulation effect implies a downstream-specific dose-response that the upstream placebo does not reproduce; an attenuation equally present upstream is siting, not regulation. Of the 21 treated matched basins, 15 have both an upstream and a downstream gauge, giving near-national coverage rather than a single case study. The contrast is also split by Köppen region to test whether a regulation effect emerges in the high-supply south. Because upstream gauges sit at higher elevations than downstream gauges, we test the placebo’s comparability assumption directly: we quantify the elevation gap and regress the per-gauge transmission slope on elevation among undammed control gauges, so that any elevation-driven regime difference is measured rather than assumed.

507 2.6 Inference

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509 With only ~21 treated clusters, cluster-robust standard errors over-reject, so the primary
510 inference is **randomization inference**: the treatment label is permuted among units
511 within Köppen $\times$ aridity- tercile strata (999 permutations) and the observed statistic
512 is compared against the resulting null distribution; cluster-robust confidence intervals
513 are reported alongside for completeness. Parallel pre-trends are checked with a dynamic
514 event study (dammed $\times$ year, reference 2009), and residual confounding is probed
515 with an aridity placebo that relabels the most-arid control tercile as pseudo-treated.
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522 2.7 Software and reproducibility

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524 All analyses are implemented in R within a **targets** pipeline (matched-set construction,
525 estimators, and figures are individual targets), using **WeightIt/cobalt** for balancing
526 and diagnostics, **fixest** for fixed-effects regression, **sf/terra/exactextractr** for the
527 spatial extractions, and **renv** for package pinning. Every result in the manuscript
528 corresponds to a named pipeline target; no figure or table is produced by hand.
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534 3 Results

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537 3.1 Reservoirs mark, but do not make, irrigated vulnerability

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539 Across the matched design (21 dammed vs 244 control subcuencas, effective control
540 sample 91 after entropy balancing), dammed basins contain roughly 4.6 times more
541 cropland than matched controls — a mean of 11.1% of basin area versus 2.4% (Figure 1,
542 panel a). This gap, however, is a *siting* signature: it is present and stable across the
543 entire 2000–2024 record, with dammed and control trajectories evolving in parallel
544 through the 2010-onset megadrought (dashed line). A dynamic event study confirms
545 this — the dammed $\times$ year coefficients on cropland area are jointly flat before 2009
546 (Wald $p = 0.68$) and show no divergence after the drought onset (Figure 1, panel
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b). Reservoirs mark where irrigated agriculture already was; they do not detectably expand it or change how it responds to drought.

3.2 No operational reservoir effect survives once siting is matched out

We probe the operational vulnerability mechanism with four independent estimators (Table 1): two cross-sectional doubly-robust matched ATTs — net irrigated-area expansion (-0.69 ha km $^{-2}$, 95% CI $[-2.4, 1.1]$) and reservoir evapotranspiration buffering (-0.014 d log ET / d SPEI, $[-0.065, 0.037]$) — and forcing-interacted difference-in-differences slope gaps on the observed area and ET panels, in which the deficit dose is the meteorological SPEI rather than calendar time (so year fixed effects absorb the common megadrought shock and the estimand is the differential deficit-to-impact *slope*). Because the treatment is effectively time-invariant (18 of 24 reservoirs predate the record), this within-unit, forcing-based design is the only defensible dynamic identification; inference uses randomization of treatment within climate $\times$ aridity strata (~ 21 treated clusters).

Under the design’s primary inference, no estimate indicates an operational effect (Figure 2, Table 1). Three of the four straddle zero outright: net irrigated-area expansion, and the slope gaps for irrigated area ($p_{\text{perm}} = 0.91$) and orchard ET ($p_{\text{perm}} = 0.24$). The whole-basin ET slope gap is the exception, and we treat it explicitly rather than fold it silently into the nulls. Its cluster-robust interval is positive and excludes zero ($+0.048$, $[0.026, 0.070]$); however, it fails the design’s pre-specified randomization inference ($p_{\text{perm}} = 0.11$) and, decisively, its dynamic event study shows strongly non-parallel pre-trends ($p \approx 10^{-10}$). The panel is therefore confounded — whole-basin MODIS ET is contaminated by residual aridity over sparsely vegetated, barren land — so neither its point estimate nor its cluster-robust interval is interpretable, and for the same reason it is the one outcome we exclude from the equivalence bounding

Reservoirs do not alter irrigated-area dynamics

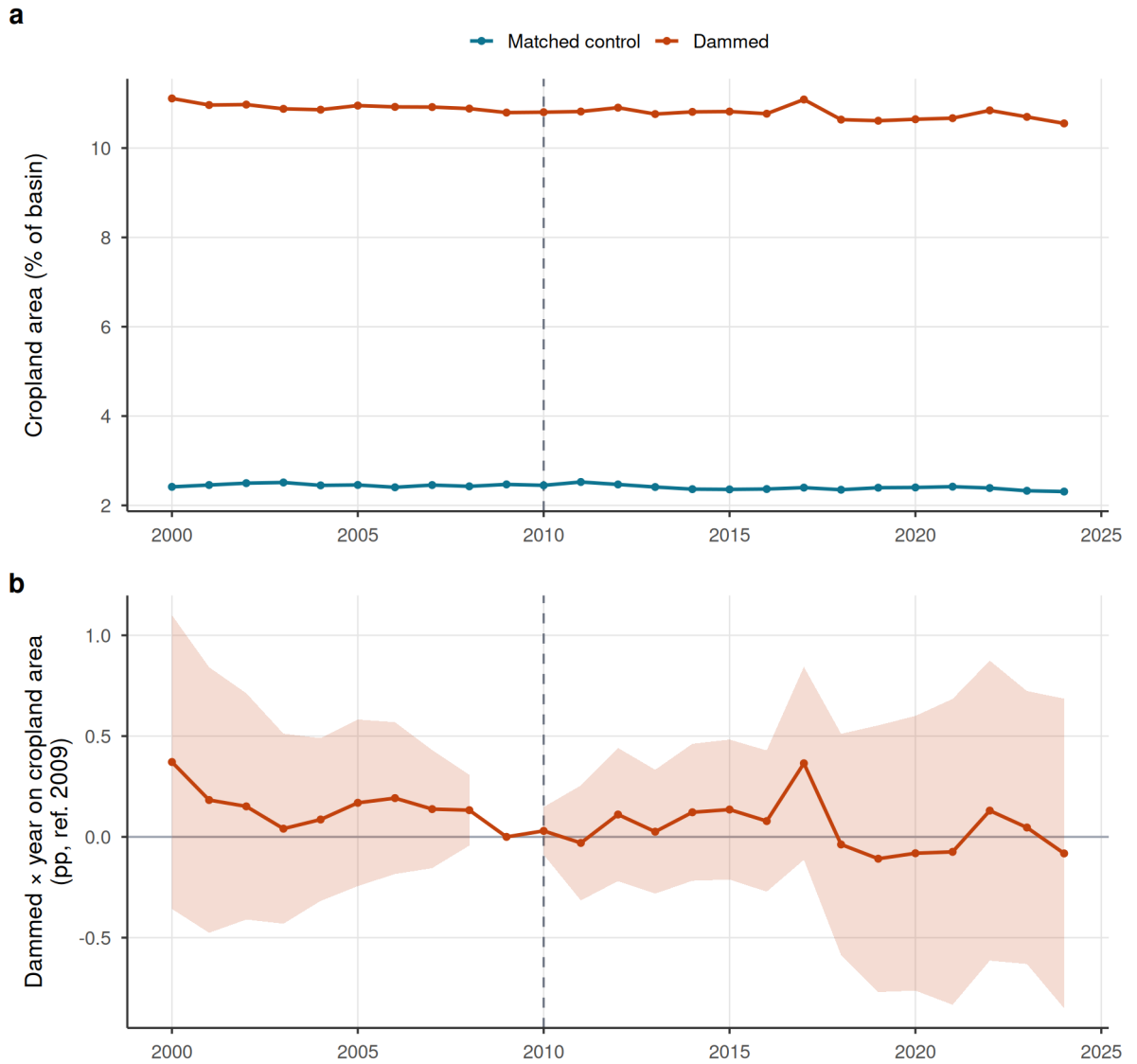


Figure 1: Observed irrigated-area dynamics in dammed versus matched-control basins. (a) Weighted mean cropland area (% of basin, MapBiomass class 18); dammed basins hold ~4.6x more cropland (11.1% vs 2.4% of basin area) but trajectories are parallel. (b) Dynamic event study (dammed x year, reference 2009): flat pre-trends and no post-megadrought divergence; shaded band is the 95% confidence interval (coefficient ± 1.96 SE). Dashed line, 2010 megadrought onset.

below. Two facts keep this exception from rescuing a buffering narrative: taken at face value the estimate points toward *vulnerability* (a steeper deficit-to-impact slope), not buffering; and the cleaner orchard-ET stratum and the cross-sectional ET-buffering ATT both straddle zero. Across the interpretable outcomes, then, no panel shows the positive slope the vulnerability mechanism predicts nor the negative slope operational buffering would produce: conditional on siting, the operational reservoir effect brackets zero.

Table 1: Reservoir-effect estimates across estimators. Cross-sectional doubly-robust matched ATTs and forcing-interacted DiD slope gaps (treat:SPEI). DiD rows are judged by randomization-inference p-values (p), the design’s primary inference at ~21 clusters; cluster-robust CIs are shown for completeness and over-reject here. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering. *Whole-basin ET fails its event-study pre-trend test (p approximately 1e-10): its cluster-robust CI excludes zero but the panel is confounded by barren-land aridity, so it is labelled confounded and excluded from the equivalence bounds (Table 2).

Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Cross-sectional matched ATT	Irrigated-area expansion	ha km ⁻² added	-0.687	[-2.44, 1.06]	-	null
Cross-sectional matched ATT	Reservoir ET buffering	d log ET / d SPEI	-0.014	[-0.0646, 0.0369]	-	null
Forcing-interacted DiD (slope-gap)	Irrigated area	differential deficit->impact slope	0.001	[-0.00344, 0.00493]	0.91	null
Forcing-interacted DiD (slope-gap)	Whole-basin ET	differential deficit->impact slope	0.048	[0.0255, 0.0703]	0.11	confounded*

Table 1: Reservoir-effect estimates across estimators. Cross-sectional doubly-robust matched ATTs and forcing-interacted DiD slope gaps (treat:SPEI). DiD rows are judged by randomization-inference p-values (p), the design’s primary inference at ~21 clusters; cluster-robust CIs are shown for completeness and over-reject here. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering. *Whole-basin ET fails its event-study pre-trend test (p approximately 1e-10): its cluster-robust CI excludes zero but the panel is confounded by barren-land aridity, so it is labelled confounded and excluded from the equivalence bounds (Table 2).

Group	Outcome	Estimand	Esti-		Ver-	
			mate	95% CI	p	dict
Forcing-interacted DiD (slope-gap)	Orchard ET	differential deficit->impact slope	- 0.020	[-0.0448, 0.00565]	0.24	null

3.3 Sensitivity to aridity adjustment

Dammed basins are systematically more arid than controls, and the matched set does not fully remove this: baseline aridity remains imbalanced after weighting (standardized mean difference 0.17, above the 0.1 target), so the doubly-robust adjustment for aridity is load-bearing rather than a cosmetic mop-up. Because dammed basins are the more arid, residual under-control biases *toward* apparent vulnerability, which makes the nulls conservative. Functional form matters and we report it: in the ecological-proxy transmission-slope analysis the dammed-versus- control estimate falls from +0.33 under linear aridity adjustment to +0.11 once a quadratic aridity term is added, and in the irrigation-specific stratum it collapses to 0 (0.08 $\rightarrow$ 0.01). The apparent steepening is therefore largely a water-limitation/aridity artifact — consistent with the siting interpretation — but the same sensitivity means estimates that rely on linear aridity control should be read as functional-form-dependent, and we treat the curvature-robust values as the conservative reference.

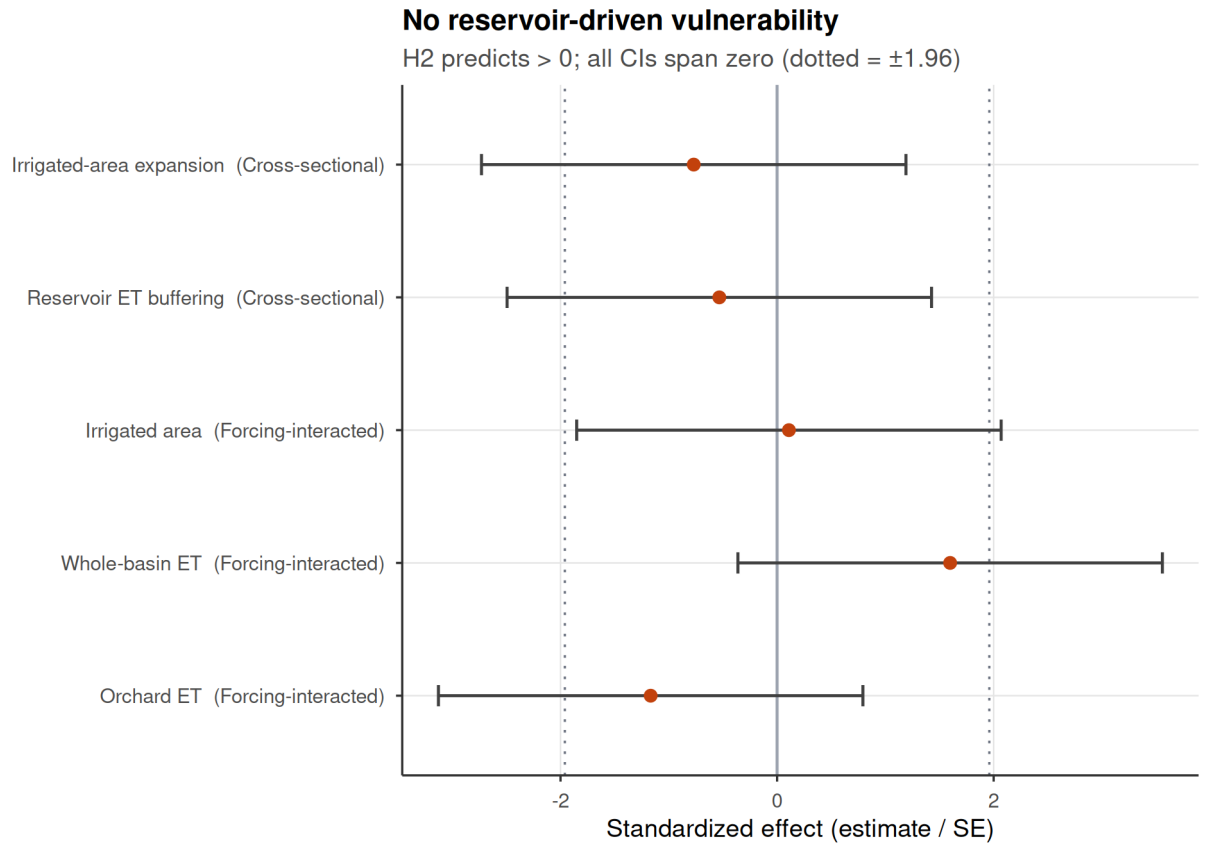


Figure 2: Standardized dammed - control effect across the four estimators. DiD rows are standardized via their randomization-inference p-value (the design's primary inference), so whole-basin ET - cluster-significant but permutation-null and pre-trend-confounded - sits within the bounds; cross-sectional ATTs use estimate/SE. Horizontal bars span ± 1.96 standardized units (the 95% reference interval); dotted lines mark ± 1.96 .

3.4 Bounding the null: what the design can rule out

A null is only informative against a stated effect size, so we convert each non-result into a bound via equivalence testing (two one-sided tests) and a minimum detectable effect (MDE) built from the permutation-null spread (Table 2). We bound the four interpretable outcomes (streamflow ITT and downstream, irrigated-area DiD, orchard-ET DiD); whole-basin ET is excluded because its pre-trends fail, so an equivalence

interval on a confounded estimate would be meaningless. The streamflow design has the most power: it can detect a buffering effect that attenuates SPEI→SSI transmission by 46% of the baseline slope (ITT MDE 0.27 against a baseline transmission of 0.59; downstream-only MDE 0.31 against its 0.53 baseline, 58%). Here “baseline” is the untreated (control) SPEI→SSI slope from the fixed-effects model (Table 2), which is distinct from — and should not be confused with — the raw per-gauge mean transmission of downstream gauges (0.599) reported for the placebo comparability check (Table 4). No outcome’s 90% interval falls inside its pre-specified negligible region ($\pm 25\%$ of baseline), so we do *not* claim formal statistical equivalence to zero — with ~21 treated clusters the design can exclude only **large** operational effects, not small ones — a policy-relevant buffer of 20–45% of baseline transmission is *not* excluded. The causal weight of the null therefore rests on the within-basin placebo below, not on equivalence: the design excludes only large operational effects, and the placebo indicates that the residual apparent attenuation is selection rather than regulation.

Table 2: Equivalence / minimum-detectable-effect bounds on the headline nulls. Observed slope gap, MDE (smallest effect detectable at 80% power from the permutation-null SD), the 90% (TOST) interval, and whether it lies within the negligible region ($\pm 25\%$ of baseline). ‘baseline’ is the untreated (control) SPEI→SSI / SPEI→impact transmission slope from the fixed-effects model. No outcome is formally equivalent to zero; the streamflow design rules out attenuations larger than ~46-58% of baseline transmission.

Outcome	Observed	MDE	90% CI	Equiv. to	
				0?	p_perm
streamflow SSI (ITT)	-0.183	0.268	[-0.34, -0.0253]	no	0.39
streamflow SSI (downstream)	-0.165	0.312	[-0.348, 0.0186]	no	0.48
irrigated area (DiD)	0.001	0.010	[-0.00506, 0.00656]	no	0.90
orchard ET (DiD)	-0.020	0.046	[-0.0463, 0.00718]	no	0.25

To confirm that this null is informative rather than merely underpowered — that the design could detect a true effect of the relevant order — we ran a positive control on the streamflow estimator (Table 3). Injecting a known buffering slope into the treated downstream units, the estimator recovers it almost exactly (a -0.30 injection yields -0.47 on top of the -0.17 baseline), the no-injection case is correctly *not* flagged ($p_{\text{perm}} = 0.48$), and randomization inference detects the effect once the total slope reaches -0.47 ($p_{\text{perm}} = 0.014$), with -0.62 strongly detected ($p_{\text{perm}} = 0.001$). The design therefore has the power to reveal a large operational buffering effect where one exists, so its failure to find one is evidence of absence at that magnitude — consistent with the MDE above — rather than blindness. This positive control validates the streamflow estimator specifically; the irrigated-area and ET nulls are not independently power-validated — their minimum detectable effects are large (for irrigated area, far above any plausible effect; Table 2) — and rest on the placebo and the conservative direction of the aridity bias rather than on their own power.

Table 3: Positive control on the streamflow estimator. A known buffering slope (beta_injected; negative = buffering) is added to treated downstream units; the estimator recovers it (recovered = baseline $-0.17 + \text{beta}$), the no-effect case is not flagged, and randomization inference detects the effect once it exceeds the minimum detectable effect.

beta_injected	recovered	p_perm	detected
0.00	-0.165	0.480	FALSE
-0.15	-0.315	0.135	FALSE
-0.30	-0.465	0.014	TRUE
-0.45	-0.615	0.001	TRUE

875 3.5 The apparent buffering is reservoir siting, not regulation

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877 The preceding nulls use vegetation and ET proxies. We obtain the same answer on the
878 direct, vegetation-independent availability outcome — streamflow — and, uniquely
879 here, can show *why* a naive analysis would mislead. From CR2 daily gauge records
880 (2000–2020) we build a 12-month Standardized Streamflow Index (SSI-12) and test
881 whether meteorological drought (SPEI-12) propagates less strongly to streamflow
882 drought in dammed than in matched-control basins (Figure 3). Dammed basins do
883 show a flatter SPEI→SSI transmission — the visual signature of buffering (Figure 3,
884 panel a). But this apparent attenuation (i) does not survive randomization inference
885 (intent-to-treat slope gap -0.18 , cluster $p = 0.03$ but $p_{\text{perm}} = 0.39$), and (ii) is
886 present just as strongly at gauges located *upstream* of the dam — which are physically
887 unregulated — as at downstream gauges (-0.20 vs -0.16 ; Figure 3, panel b). There
888 is no downstream-versus-upstream dose-response, the signature a genuine regulation
889 effect would require. The flatter slope is therefore a property of *where* reservoirs are
890 sited — arid, hydrogeologically distinct basins — not of the reservoirs themselves.

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901 The siting interpretation is confirmed by the way the signal behaves across the
902 aridity gradient. The national “buffering” slope gap (-0.16 to -0.20) **collapses to**
903 -0.03 **once the comparison is made within climate region** — the textbook
904 signature of a between-region confound, in which arid treated basins are contrasted
905 against a control pool spanning the whole wet→dry gradient. In the high-supply
906 south, where buffering should be easiest to detect, the downstream and upstream gaps
907 are statistically identical (-0.036 vs -0.034) and both indistinguishable from zero
908 (southern downstream $p_{\text{perm}} = 0.78$). Splitting by region thus both kills the signal and
909 equalizes the upstream and downstream reaches — two independent confirmations that
910 the attenuation is the aridity gradient, not regulation. Because the upstream-of-dam
911 comparison is unavailable for any vegetation outcome, streamflow provides the cleanest
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within-design falsification: the effect a naive dammed-versus-control or calendar-time analysis would report is siting, not regulation.

The placebo assumes upstream and downstream reaches are hydrologically comparable, which we test directly (Table 4). Upstream gauges do sit higher than downstream ones (mean 1.0 vs 0.46 km), and gauge elevation is not innocuous — among undammed control gauges the SPEI→SSI transmission slope declines by 0.21 per kilometre of elevation ($p < 0.001$). The placebo nonetheless holds, for two reasons. First, the *raw* transmission slopes of upstream and downstream treated gauges are nearly identical (0.565 vs 0.599), so the elevation difference does not produce a transmission-regime gap of the size at issue. Second, the apparent buffering is if anything *stronger* upstream (-0.20) than downstream (-0.16) — the opposite of what regulation, which acts only downstream, would produce. We cannot claim the reaches are perfectly interchangeable, and a residual elevation/regime component remains a caveat, but neither the raw slopes nor the direction of the gap supports a regulation effect masked by elevation.

Table 4: Upstream/downstream placebo comparability. Upstream gauges are higher than downstream, and gauge elevation predicts the transmission slope among undammed controls (-0.21 per $+1000$ m), yet upstream and downstream treated gauges have nearly identical raw transmission slopes - so the equal up/down attenuation is not an elevation artifact.

quantity	value	detail
upstream gauges (elev, mean m)	1003.000	n=28; mean transmission 0.565
downstream gauges (elev, mean m)	459.000	n=40; mean transmission 0.599
control transmission slope vs elevation (per $+1000$ m)	- 0.212	p = 0.00 (n=84 control gauges); $\sim 0 \Rightarrow$ elevation does not drive transmission

Apparent streamflow buffering is siting, not the reservoir

Effect is permutation-null and as strong in UPSTREAM (unregulated) gauges

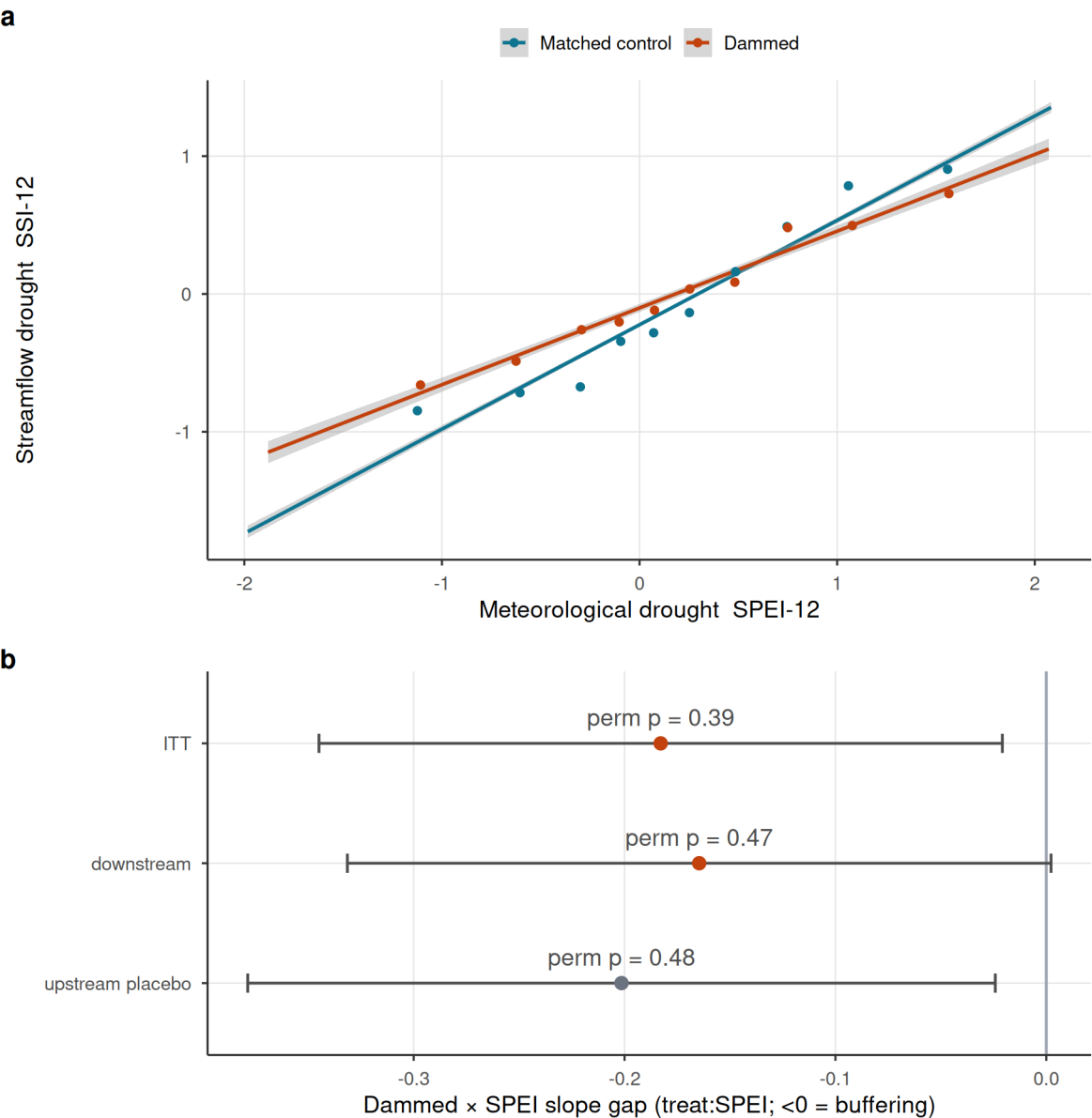


Figure 3: Streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); lines are weighted least-squares fits and shaded bands are 95% confidence intervals; dammed basins show a flatter slope, the apparent buffering. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo; horizontal bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE), annotated with randomization-inference p-values. All are permutation-null, and the upstream placebo - above-dam, unregulated - is as strong as downstream, so the attenuation is siting/aridity, not the dam.

3.6 Storage declines without a loss of buffering capacity

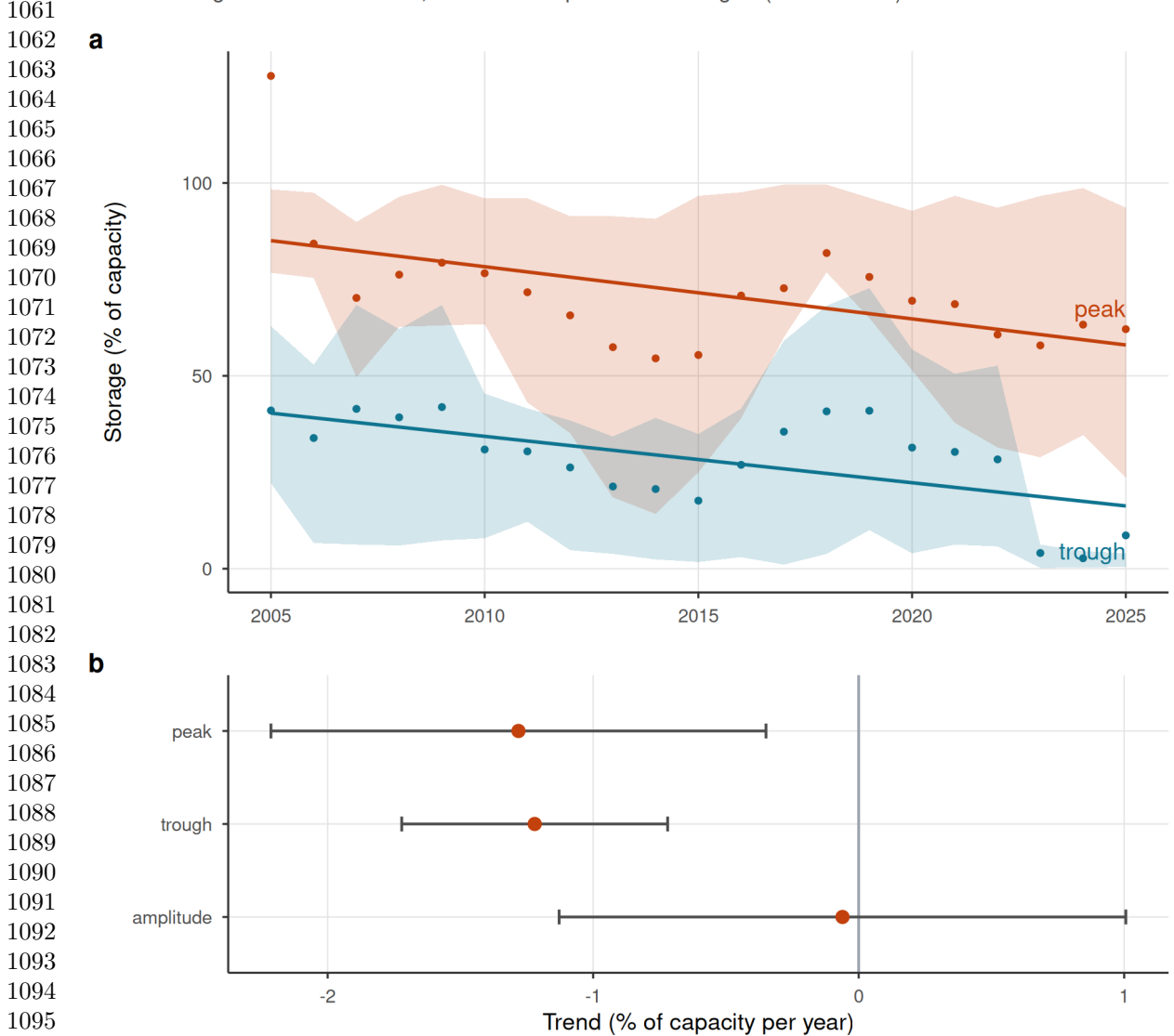
If reservoirs neither buffer nor amplify drought operationally, the relevant change under the megadrought is in how much water is available to store (Figure 4). Over 2005–2024 the entire storage band drifts downward in step: annual peak storage declines by -1.3 percentage points of capacity per year ($p = 0.013$) and the annual trough falls almost identically (-1.2 pp yr⁻¹, $p < 10^{-4}$), while the seasonal amplitude is statistically unchanged (-0.06 pp yr⁻¹, $p = 0.91$). A band that drops without narrowing is consistent with a supply-side level shift rather than a degraded buffer, which would compress the amplitude; reservoirs still convert meteorological surplus into storage as efficiently as before. We caution that this storage trend is descriptive — a pooled within-reservoir trend with no control group or naturalized-inflow series — so it cannot by itself distinguish declining inflow from rising demand-side drawdown. The most direct demand proxy available to us weighs against the demand explanation, however: irrigated area in dammed basins shows no expansion over the period (stable, parallel trajectories, Figure 1; differential-expansion ATT indistinguishable from zero, Table 1), so a growing consumptive footprint is not evident as a driver of the storage decline — though a compositional shift toward more water-intensive perennials could raise per-area use without enlarging area. Read alongside Chile’s documented aridification, the pattern is therefore more consistent with a falling water supply than with any loss of storage or buffering capacity, and on that reading the operational case for expanding storage as drought protection is weak^{2;25}.

4 Discussion

Across a matched national design and four independent estimators — cross-sectional irrigated-area expansion and ET buffering, and forcing-interacted slope gaps on observed area and ET — the operational effect of reservoirs on drought outcomes brackets zero once siting is matched out. The same answer holds for the direct,

1059 **The binding constraint is inflow, not storage**

1060 Whole storage band shifts down; seasonal amplitude unchanged (buffer intact)



1097 **Figure 4:** Storage declines without a loss of buffering capacity. (a) Cross-reservoir
 1098 annual storage band - peak and trough percent-of-capacity (mean line; shaded bands are
 1099 the interquartile range across reservoirs) - drifts downward over 2005-2024 with roughly
 1100 parallel trend lines: the whole band shifts while seasonal amplitude is unchanged. (b)
 1101 Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects); error bars
 1102 are 95% confidence intervals (estimate $\pm$ 1.96 cluster-robust SE): peak and trough
 1103 both decline while amplitude brackets zero, consistent with a supply-side level decline
 1104 rather than a refill/buffer degradation. Descriptive: pooled within-reservoir trend, no
 control group.

vegetation-independent availability outcome: the apparent flattening of meteorological-
to-streamflow drought propagation in dammed basins is as strong at physically
unregulated *upstream* reaches as downstream, and collapses from a slope gap of -0.18
to -0.03 when the comparison is made within climate region. Dammed basins do hold
roughly 4.6 times more cropland than matched controls (11.1% vs 2.4% of basin area),
but this gap is fixed across the record, with parallel pre-trends and no divergence
through the megadrought. Taken together, the evidence supports a single interpretation:
reservoirs **mark** where water-dependent vulnerability already concentrated
— arid, already-irrigated basins — rather than **make** it. The vulnerability signature
is a property of siting, set before the drought and stable through it, not a dynamic
consequence of regulation.

This reframes “no effect” as the finding rather than a failure of power, but only because
a credible *cause* of the apparent effect is identified. The within-basin upstream/down-
stream placebo is the cleanest within-design falsification here: a genuine regulation
effect must appear downstream of the dam and be absent upstream, and it is not.
What remains, once that placebo and within-region conditioning are applied, is the
aridity gradient itself. The methodological corollary is general. A naive dammed-
control contrast, or a calendar-time difference-in-differences, would read the large
cross-sectional area, ET, and transmission gaps — amplified during the drought —
as a reservoir effect; conditioning on realized meteorological forcing and estimating a
deficit-to-impact *slope*, and benchmarking against an unregulated within-basin reach,
makes the signal disappear. Reservoir-buffering studies that compare dammed against
undammed basins, or pre- against post-construction levels, inherit the full siting con-
found^{7;8;11;40;41}; the upstream placebo is a transferable identification tool applicable
to any regulated river with gauges above and below the structure.

1151 If reservoirs neither buffer nor amplify drought operationally, the relevant change under
 1152
 1153 a multi-decadal dry spell is in the water available to store rather than in the structures,
 1154 and the storage record is suggestive on this point. The seasonal refill remains efficient
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 1156 and its amplitude is unchanged, yet the entire storage band drifts downward, peak and
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 1158 trough declining in step at roughly -1.3 and -1.2 percentage points of capacity per
 1159 year. A band that drops without narrowing is consistent with a falling supply ceiling
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 1161 rather than a degraded buffer. We stress that this is a descriptive, control-free trend that
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 1163 cannot on its own separate declining inflow from rising demand-side drawdown⁵²; both
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 1165 would shift the band downward without narrowing it. Our most direct demand proxy,
 1166 however, weighs against the demand explanation: irrigated area in dammed basins did
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 1168 not expand over the period (stable, parallel trajectories; a differential-expansion ATT
 1169 indistinguishable from zero), so rising consumptive demand is not evident as the driver
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 1171 — even though a crop-composition shift toward more water-intensive perennials could
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 1173 raise per-area use without enlarging area. Read against Chile’s documented transition
 1174 from episodic drought toward structural aridification — in which declining precipita-
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 1176 tion and rising atmospheric evaporative demand lower inflows independently of any
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 1178 infrastructure^{14;24;29;30;53}, echoing aridification trajectories documented elsewhere⁵⁴
 1179 — and the global pattern in which reservoirs increasingly fail to refill and the storage
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 1181 returns of new dam construction are diminishing^{2;25}, the more likely reading is that
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 1183 the binding constraint is shifting toward water supply rather than storage capacity or
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 1185 operation. A naturalized-inflow series and a demand-side decomposition are needed to
 1186 settle this, and we treat it as a strong hypothesis rather than an identified result.
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 1188 The policy implication is direct and uncomfortable for the prevailing response to
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 1190 scarcity in Chile and elsewhere. Because the 1981 Water Code makes existing water-
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 1192 use rights perpetual private property that the state cannot readily curtail during
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 1194 drought, the reflexive response has been supply-side — a national programme of new
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 1196 dams and inter-basin transfers^{15;33;35}. Our results imply that adding storage to basins

whose inflow is declining is a supply-blind response to a supply problem: it cannot
 manufacture water that aridification is removing, and the buffering benefit it is built
 to provide is, on this evidence, attributable to where dams sit rather than to what they
 do. This is the drought-domain expression of the reservoir-effect and supply-demand-
 cycle hypotheses¹, whose practical force has been contested for want of quantification³;
 we provide a credibly identified test and find the protective function over-attributed.
 The leverage instead lies on the demand and allocation side — curbing the over-
 allocation and induced expansion that the efficiency paradox entrenches^{12;13;34} — and
 in managed recharge and reallocation rather than new impoundments^{36;37;55}.

Several limitations qualify these conclusions. First, treatment is effectively time-
 invariant over the record (most reservoirs predate it), so we identify differential drought
response rather than the effect of switching a reservoir on; a staggered-commissioning
 design is infeasible here and would be the cleaner counterfactual where construction
 dates and longer records permit. Second, the result is a bounded null, not a proven
 equivalence: with ~21 treated clusters the design rules out buffering larger than 46–58%
 of baseline transmission for streamflow but cannot exclude small operational effects, so
 the causal weight rests on the placebo, not on equivalence. Third, the observed-area
 outcome (MapBiomass class 18) captures total cropland rather than irrigation status,
 and Chile’s agricultural change has been compositional — a shift toward perennial
 orchards within roughly constant cropland area — so a latent water-demand signal not
 visible in net area remains possible; whole-basin MODIS ET is additionally confounded
 by residual aridity over barren land, so the ET conclusion leans on the cross-sectional
 buffering null and the orchard stratum — both low-powered (minimum detectable
 effects of roughly twice their baselines), so for ET specifically the finding is “no *large*
 effect detected” rather than no effect. Relatedly, the matched set leaves baseline aridity
 imbalanced (SMD 0.17), so the doubly-robust adjustment for it is load-bearing and
 the estimates are sensitive to its functional form: a quadratic aridity term moves the

1243 proxy transmission-slope estimate from +0.33 to +0.11 — toward the null we report,
 1244 and a reminder that the linear-adjustment values are not the conservative ones. Fourth,
 1245 the streamflow placebo covers 15 of 21 treated basins over 2000–2020, and the high-
 1246 supply southern subset is small; a modest regulation effect in wetter basins cannot be
 1247 excluded, though drought is also least binding there. Upstream and downstream gauges
 1248 are also not perfectly interchangeable — upstream gauges sit higher, and elevation
 1249 predicts the transmission slope among controls — so a residual regime component
 1250 remains; that said, the raw upstream and downstream transmission slopes are nearly
 1251 identical and the apparent buffering is if anything stronger upstream, neither of which
 1252 is consistent with a regulation effect obscured by elevation. The 12-month aggregation
 1253 is a further boundary: it targets annual-to-interannual transmission and does not
 1254 resolve the sub-seasonal (1–3 month) operational buffering reservoirs are designed to
 1255 provide, so our conclusions concern long-term vulnerability rather than short-term
 1256 operation. Generalizability is also bounded by governance: Chile’s 1981 Water Code
 1257 is an unusually extreme, market-based regime (perpetual, tradable private rights the
 1258 state cannot readily curtail), so while the *operational* finding — that buffering reflects
 1259 siting, not regulation — should hold wherever dam siting is non-random, the demand-
 1260 feedback and policy inferences are conditioned on this weak adaptive feedback; under
 1261 state-managed or dynamically regulated allocation, operators able to curtail demand
 1262 or re-time releases could in principle produce genuine buffering, or weaker induced-
 1263 demand lock-in, than we observe here. Finally, residual SUTVA spillovers (adjacency,
 1264 shared aquifers) survive the contamination rule, and our estimand is the *drought* effect
 1265 of reservoirs — not their value for hydropower, flood control, or urban supply, which
 1266 this study does not assess.
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 1268 One alternative interpretation deserves emphasis because mean-based estimators
 1269 cannot see it: damming might leave the *magnitude* of drought impact unchanged
 1270 while recomposing its **dependence structure**, weakening the meteorology→outcome
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coupling and substituting a storage→outcome coupling — a regime shift that would itself explain the convergent null. We cannot adjudicate this with the present data: the storage-recovery dynamics are stationary (refill efficiency and amplitude are flat), which weakens the premise, and the decisive test requires reservoir release and operating-rule records that set storage *independently* of contemporaneous inflow. The only storage signal on disk is monthly change in storage (a drawdown proxy), which is an endogenous hydrological response to the same precipitation and inflow forcing — and partly to demand — and is therefore collinear with SPEI; regressing an outcome on it cannot separate a genuine storage→outcome regime from the forcing→outcome coupling we already estimate, because it carries no independent allocation information. Identifying the dependence-shift limb thus needs exogenous release or rule-curve data. Absent that, and given that any apparent decoupling must survive the upstream placebo, the dependence-shift account reduces to the siting interpretation. Resolving it — together with longer streamflow records and water-rights allocation data — is the priority for future work.

Returning to the question that motivates the study — do reservoirs create resilience or merely postpone vulnerability under prolonged drought? — our evidence indicates that, under aridification, they do neither operationally: storage neither buffers nor amplifies the drought signal beyond what basin siting already implies, while the water available to store steadily declines. Resilience to a falling supply ceiling will not come from more impoundments but from confronting demand and allocation — the harder, and unavoidable, adaptation.

5 Data and code availability

The full analysis pipeline — the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table — is openly available at https://github.com/frzambra/dam_drought_

effectiveness and archived on Zenodo (DOI: 10.5281/zenodo.XXXXXXX). The pinned R environment (`renv.lock`) reproduces the package versions used. Reservoir storage records, the derived result tables, and matched-set metadata are included in the repository; the third-party input datasets are publicly available from their providers as cited: CHIRPS/CHIRTS (standardized drought indices), MODIS MOD16/MOD13/MOD17 (evapotranspiration and vegetation), MapBiomass Chile (land cover), the CR2 network (streamflow), SRTM (elevation), and the DGA reservoir and watershed registries. Every result in the manuscript corresponds to a named pipeline target, so the analyses can be regenerated with a single `targets::tar_make()`.

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